Navy Paint Analysis MVP Feature-Gap Analysis

Assessment Status

Field	Value
Status	Updated against the time-boxed MVP requirements; implementation not performed
Assessed	2026-08-30 (America/Chicago)
Requirements source	`docs/Architecture/MVP_Product_Requirements.md`
Implementation assessed	`Z:\mvp20-split` working tree
Delivery boundary	One project, one active campaign, one controlled deployment, one to two weeks
Product objective	Human-led incumbent characterization plus one bounded candidate/lab-feedback decision
Readiness conclusion	Reusable trust and optimization primitives exist, but the integrated Navy workflow is largely missing

This is an implementation-gap analysis, not a scientific finding, qualification assessment, or authorization to test or release a coating.

Executive Conclusion

The latest MVP is much smaller than the earlier naval roadmap, but it is not already present in the codebase. The repository is primarily a collection of Rust/Python command-line modules, schemas, fixtures, evidence packages, and orchestration controls. It does not contain the required integrated project workspace, transactional application data model, seven-screen experience, or end-to-end laboratory feedback workflow.

The strongest reusable assets are:

- strict candidate and score envelopes;
- stable hashes, replay, provenance, and authority controls;
- formulation arithmetic and feasibility-oriented modules;
- CRO protocol and capture-template packaging;
- evidence-intake validation patterns;
- optimizer constraint and ranking boundaries;
- report packaging and claim-limiting patterns.

The largest gaps are:

- project, user, role, sample, custody, test, observation, hypothesis, and review persistence;
- protected file upload and evidence-to-record linking;
- a usable operator/scientist interface;
- the Navy-specific CSV and protocol/feedback contracts;
- the LLM adapter and typed, human-reviewed exchanges;
- a small formulation mutation/selection runner joined to accepted laboratory feedback;
- deterministic fitness over approved `GeneticFeedback`;
- application audit, backup/restore, and Markdown/PDF report assembly.

Planning posture:

- **Underlying reusable primitives:** moderate; approximately 40–50% of the needed technical concepts have implementation or contract precedent.
- **Integrated MVP workflow:** low; approximately 15–25% is directly runnable as the required product.
- **User-facing acceptance criteria:** very low; the current modules do not complete the required Navy project end to end.

These ranges are planning estimates, not earned-value metrics. A one-to-two-week delivery is plausible only with aggressive reuse, operator-entered configuration, a simple local deployment, and representative lab-return fixtures if physical testing does not return in time.

Scope Boundary

This MVP does not require:

- automatic instrument interpretation or proprietary parsers;
- automatic identification of the original recipe;
- a coatings knowledge refinery or neuro-symbolic engine;
- cross-project learning;
- autonomous agents or laboratory coordination;
- multiple simultaneous customers;
- enterprise IAM, multi-tenancy, queues, HA, billing, or integrations;
- performance improvement, production release, naval qualification, or manufacturing authority.

Those items must not expand the one-to-two-week build.

Current Implementation Baseline

Reusable without conceptual redesign

Existing capability	Evidence in repository	MVP use
Strict candidate contract	`amphora-candidate-contract`	Base identity and typed candidate payloads
Stable hashing and deterministic serialization	`amphora-core` and candidate-contract utilities	File, request, result, configuration, and replay fingerprints
Run provenance and replay patterns	`amphora-run-provenance` and CLI campaign controls	Genetic seed/version/replay records
Formulation arithmetic	`amphora-formulation-arithmetic`	Totals, range validation, and residual calculation after adaptation
Evidence validation	`amphora-evidence-intake-validator`	Fail-closed intake and disposition patterns
CRO packet and return template	`amphora-cro-protocol-packager`	Starting point for one Navy protocol and return CSV
Constraint/ranking governance	`amphora-optimizer-orchestrator`	Deterministic hard gates and fitness authority boundary
Human deliverable packaging	`amphora-report-packager`	Starting point for report and manifest
CLI orchestration	`amphora-cli`	Operator-run fallback and integration shell

These are foundations, not completed MVP features. Most operate on module envelopes and files rather than the required application records and review lifecycle.

Partial or mismatched foundations

Area	Current posture	Gap
Candidate lineage	Hashes and candidate identities are well represented	No Navy formulation/campaign workflow joining candidates to tests and feedback
Evidence	Candidate-bound packages and validation rules exist	No observation ledger with source locations, review states, and corrections
Custody	CRO templates contain some custody/raw-file fields	No sample, derived-sample, or custody-event model and UI
Protocol packaging	CRO packager and capture template exist	No Navy protocol bound to controls, replicates, candidate versions, and return fields
Optimization	Constraint/ranking orchestration exists	No small numeric mutation loop, campaign persistence, or lab-feedback next-generation decision
Reporting	Report artifacts and claim controls exist	No project report assembling custody, evidence, hypotheses, mass balance, and approvals
Operations	CLI and control-plane patterns exist	No protected web workspace, role workflow, backup/restore proof, or application errors

Missing as an integrated capability

- authenticated project workspace and role assignments;
- durable project database and protected project file store;
- sample registry and custody timeline;
- test plan and result-package lifecycle;
- observation CSV import, correction history, and evidence review;
- composition hypotheses, support/conflict links, and distinct mass-balance bases;
- readiness decision and approval workflow;
- configured external LLM exchange with typed validation and human approval;
- lab-report extraction with source locations, confidence, discrepancies, and missing-output flags;
- approved `GeneticFeedback`, deterministic fitness, and next-generation/stop decision;
- seven minimal product screens;
- application audit export and tested database/file restore.

No production FastAPI/React application matching `docs/Architecture/Implementation_Architecture.md` was found. That document is a target design, not evidence of implemented routes, tables, or screens.

Requirement-by-Requirement Gap

Status meanings: **Reusable** means directly useful implementation exists; **Partial** means related code or a contract exists but does not satisfy the requirement; **Missing** means no supported implementation was found; **Manual** is intentionally outside the software.

Project, people, samples, and files

Requirement	Status	Required MVP work
One project with team, status, questions, limitations, and documents	Missing	Add project/membership records and document upload
Named operator, scientist, reviewer, and program-lead actions	Missing	Add authentication, roles, and named approvals
Sample types, lots, condition, quantity, storage, photos, and notes	Missing	Add sample records and file associations

Requirement	Status	Required MVP work
Derived-sample parentage	Partial	Reuse lineage concepts in a simpler sample relationship
Custody events	Partial	Turn CRO custody fields into an append-only sample timeline
Protected original files with SHA-256	Partial	Reuse hash utilities; add storage, access checks, and download

Tests, observations, and review

Requirement	Status	Required MVP work
Manual test plan and outcome states	Missing	Add test and result-package records/forms
Versioned protocol bound to candidates, controls, replicates, and process	Partial	Specialize the CRO packager and persist approval/version
One generic return CSV	Partial	Reduce the capture template to one documented and tested contract
Return completeness and identity checks	Partial	Adapt evidence intake for files, columns, units, controls, replicates, deviations, and hashes
Manual observation entry and CSV import	Missing	Add preview, row validation, visible errors, and editable drafts
Immutable originals and correction history	Missing	Add observation revisions rather than destructive overwrite
Evidence and fitness dispositions	Missing	Add reviewer workflow and eligibility enforcement
Source-file and page/table/row references	Missing	Add source locators to observations and extracted fields

Composition and decision

Requirement	Status	Required MVP work
Human component/role hypotheses and ranges	Missing	Add hypothesis editor and review state
Supporting and conflicting observations	Missing	Add explicit evidence links
Alternatives, unknowns, assumptions, and limitations	Missing	Persist without forcing identity resolution
Mass balance by declared basis	Partial	Adapt arithmetic for min/best/max totals and unidentified residual
Invalid-value, unit, basis, and total warnings	Partial	Add visible blocking/warning states
Reconstruction-readiness decision	Missing	Add allowed decisions, rationale, blockers, reviewer, and time

LLM communication

Requirement	Status	Required MVP work
One configured LLM adapter	Missing	Implement one provider; keep secrets outside source control
Typed `GeneticRunRequest` draft	Missing	Define schema, assemble approved context, parse strictly, require approval
Typed lab-return extraction	Missing	Include identity, endpoints, units, deviations, failures, missingness, confidence, and source

Requirement	Status	Required MVP work
Typed `GeneticFeedback` draft	Missing	Map validated fields into a scientist-editable draft
Exchange audit	Missing	Persist request, response, context, model, validation, reviewer, and cost where available
Safety boundary	Missing	Prevent free-form/malformed output from launching runs, accepting evidence, or calculating fitness

Existing Claude-oriented controller material is engineering/campaign automation, not an in-product Navy scientific LLM adapter. It does not satisfy these requirements.

Genetic proposal and lab feedback

Requirement	Status	Required MVP work
One formulation genome	Partial	Define a typed list of existing ingredient/loading and optional process variables
Mutable/fixed flags, bounds, and precision	Partial	Add one reviewed configuration schema
One numeric mutation strategy	Missing	Implement deterministic bounded perturbation; no recombination
Simple parent selection	Partial	Reuse selection concepts but implement one explicit MVP rule
Feasibility checks	Partial	Adapt arithmetic for totals, negatives, ranges, and basis
One objective profile	Partial	Define hard/soft/informational endpoints, weights, units, uncertainty, and missing-data rules
Complete attempt and lineage ledger	Partial	Reuse candidate/provenance patterns in one campaign store
Approved feedback eligibility	Missing	Require evidence references and scientist approval before fitness
Deterministic fitness	Partial	Reuse optimizer boundaries; implement transparent calculation outside LLM
Next-generation or stop	Missing	Add reviewer decision and pause/stop rules
Replay from seed/versioned inputs	Reusable/partial	Reuse hashing/provenance and test full campaign replay

The current optimizer is not itself the required genetic loop. It consumes governed score envelopes and protects ranking authority, but the MVP still needs a formulation mutation operator, small campaign model, and explicit lab-return-to-fitness bridge.

UI, audit, report, and operations

Requirement	Status	Required MVP work
Seven minimal screens	Missing	Build compact forms/tables; routes may combine panels
Actor/time/action/reason audit	Partial	Extend provenance to CRUD, review, LLM, run, report, and admin events
End-to-end conclusion trace	Missing	Join conclusion through evidence, source, protocol, candidate/sample, custody, and lineage
Markdown/PDF report	Partial	Assemble Navy sections from records and permit scientist editing
Evidence manifest and hashes	Partial	Assemble a project-scoped manifest using existing hashing

Requirement	Status	Required MVP work
Backup and restore	Missing	Add simple database/file backup and perform one restore test
Visible failures without accepted partial data	Missing	Add transactional imports and error visibility

Acceptance-Criteria Coverage

AC	Outcome	Coverage
1	Project and users	Missing
2	Samples and custody	Missing/partial primitives
3	Test plan and outcomes	Missing
4	Reports, raw files, and metadata	Partial primitives
5	Source-bound observations	Missing
6	Evidence dispositions	Missing
7	Hypotheses, conflicts, and unknowns	Missing
8	Transparent mass balance	Partial arithmetic
9	LLM draft → approval → proposal	Missing integration
10	Screening batch and controlled protocol	Partial packager
11	Validated lab return	Partial validator
12	Extraction → correction → approved feedback	Missing
13	Deterministic approved-feedback fitness	Partial optimizer foundation
14	Human next-generation/stop and retention	Partial lineage foundation
15	Reproducible proposal and selection	Partial replay foundation
16	Malformed LLM output cannot gain authority	Missing in-product enforcement
17	LLM/genetic output separated from measured fact	Partial claim controls
18	Reconstruction-readiness decision	Missing
19	Approved report and manifest	Partial packaging foundation
20	End-to-end trace	Missing
21	Hash-preserving restore	Missing

None of the 21 criteria is currently satisfied end to end as a supported Navy workflow. Several have strong reusable components, making a focused integration effort feasible.

Minimum Build Needed

Must complete

1. **Thin application shell:** one deployment, local authentication/four named roles, transactional database, protected project files, and simple server-rendered pages or a minimal SPA/API.
2. **Project evidence spine:** project, membership, sample, custody, test, result package, file, observation/revision, evidence disposition, hypothesis/evidence link, decision, and audit event; hash every upload; provide one fail-closed CSV importer.
3. **Human scientific workflow:** sample/test/file/observation forms, hypotheses and evidence links, mass balance, readiness decision, report, PDF, and manifest.
4. **Bounded proposal loop:** one genome, seeded mutation, feasibility gates, simple selection, complete attempt lineage, and one objective profile.

5. **Lab feedback bridge:** one protocol/return CSV, deterministic validation, one LLM adapter, scientist correction/disposition, approved feedback, deterministic fitness, and human next-generation/stop.
6. **Operational proof:** audit export, malformed-input and authority-boundary tests, replay test, backup/restore test, and representative Navy fixture if real lab data is unavailable.

Explicit shortcuts

- One hard-coded or configuration-driven project type.
- One formulation JSON shape and return CSV shape.
- Operator-entered JSON for variables, objectives, and protocol details.
- One LLM provider and prompt/version pair.
- Synchronous generation with a small population.
- One weighted deterministic fitness formula plus hard gates.
- Local project-scoped protected file storage.
- Simple role matrix instead of generalized policy administration.
- Incumbent or scientist-approved baseline as the only control type.
- Fixture data if laboratory turnaround exceeds the sprint.

Cut first if time is lost

- Merge UI screens while preserving all required tasks.
- Make optional LLM result explanation a downloadable draft.
- Use operator-assisted report layout cleanup.
- If reliable PDF extraction cannot be completed, make structured CSV authoritative for numeric intake, retain the uploaded report, and record report reconciliation as manual.
- Support exactly one next-generation decision.

Do not cut human approval, source references, missing-data visibility, deterministic fitness, identity checks, controls, audit records, or replay inputs. They are the minimum trust boundary.

One-to-Two-Week Critical Path

Days 1–2: Contracts and storage

- Freeze project, observation CSV, protocol, extraction, `GeneticRunRequest`, `GeneticFeedback`, objective-profile, and audit schemas.
- Implement migrations and project file storage.
- Adapt candidate IDs, SHA-256, and canonical serialization.

Days 3–5: Characterization slice

- Implement users/project, samples/custody, tests/files, observations/import/review, hypotheses, and mass balance.
- Establish minimal forms/tables.
- Verify corrections, rejected evidence, unknown fractions, and incompatible bases remain visible.

Days 5–7: Proposal and protocol

- Implement genome, seeded mutation, gates, selection, lineage, and objective profile.
- Adapt CRO packaging to one Navy protocol and return template.
- Retain attempted, invalid, duplicate, selected, and untested candidates.

Days 7–9: Lab return and LLM

- Validate identity, files, columns, units, controls, replicates, deviations, and hashes.
- Implement the LLM adapter and strict typed parsing.
- Implement scientist correction/disposition and approved feedback.
- Calculate deterministic fitness and record next-generation/stop.

Days 9–10: Report and proof

- Assemble Markdown/PDF report, manifest, and audit export.
- Run malformed-input, authority-boundary, replay, and backup/restore tests.
- Complete the representative end-to-end acceptance run.

For a two-week window, remaining days should be reserved for integration defects, scientist review, deployment, and handoff. Candidate manufacture and laboratory testing are external schedule dependencies.

Principal Risks

Risk	Effect	MVP response
Schemas are not frozen immediately	Rework consumes sprint	Decide one protocol, CSV, genome, and objective profile before UI work
Existing modules are treated as drop-in services	Integration delays	Reuse algorithms/contracts selectively behind MVP adapters
Real lab turnaround exceeds build	Physical loop cannot finish in sprint	Use a clearly labeled representative fixture for software acceptance
PDF extraction is unreliable	LLM reconciliation becomes brittle	Make CSV authoritative and permit manual report reconciliation
LLM output is trusted directly	Invalid fitness/fabricated evidence	Strict parse, source binding, scientist approval, deterministic fitness
UI consumes schedule	Core workflow remains incomplete	Prefer compact forms/tables and operator-entered configuration
Backup/audit is deferred	Acceptance fails late	Implement audit and restore from the first data model
Scientific decisions remain unresolved	Objectives/tests cannot be encoded	Named scientists must supply methods, controls, ranges, and fitness policy

Decisions Required Before Coding

- Which incumbent samples and component states are available?
- What is the single MVP formulation representation?
- Which fields and units are mandatory in the return CSV?
- Which control, replication, and stop/pause rules are acceptable?
- What are the mutable variables, bounds, steps, fixed variables, and mass-balance basis?
- What is the objective profile, including hard gates, weights, uncertainty, and missing-data rules?
- Which LLM provider may receive project data, and what may leave the deployment?
- Is PDF extraction required, or may the LLM use reviewed report text plus authoritative CSV?
- Who may approve evidence, launch runs, approve feedback, decide selection, and release reports?
- Which fixture will be used if real lab feedback is unavailable?

MVP Completion Gate

The MVP is complete only when one representative Navy project demonstrates this trace without a developer editing authoritative records:

```
project and named users
  -> sample and custody
  -> test plan and uploaded files
  -> source-bound observations and review
  -> hypotheses, unknowns, and mass balance
  -> approved LLM-drafted run request
  -> reproducible bounded candidate batch
  -> selected candidates and versioned protocol
  -> validated lab return
  -> LLM extraction draft
  -> scientist correction and evidence disposition
  -> approved GeneticFeedback
  -> deterministic fitness
  -> human next-generation or stop
  -> readiness decision and approved report
  -> audit/evidence manifest
  -> successful restore with matching hashes
```

`INSUFFICIENT_EVIDENCE` and `MORE_TESTING_RECOMMENDED` remain successful outcomes. The MVP proves a controlled, reproducible evidence and decision loop; it does not prove exact reconstruction or naval qualification.

Bottom Line

The framework is useful as a trust and computation kernel, but this MVP is principally an integration and workflow build. The practical gap is not a new scientific AI platform; it is a thin, durable evidence application joining samples, files, human review, one constrained genetic proposal, one LLM-assisted lab-return review, deterministic fitness, and a defensible report.

The one-to-two-week target is credible only if scope stays at that boundary. Generalized agent orchestration, neuro-symbolic inference, instrument integrations, multi-customer support, or a polished enterprise UI would put delivery at material risk.